Client Requirements

- Customer name
- Customer address
- Customer phone number
- Number of pizzas ordered (Min 1, Max 2)
- Size of pizza (Small, Medium, Large)
- The number of extra toppings for each pizza (1= £0.75, 2 = £1.35, 3 = £2.00, 4+ = £2.00)
- Whether delivery is required
- Output total before discount and delivery
- Output discount if applied (10% over £20 excluding delivery cost)
- Output delivery charge if needed (£2.50)
- Final Cost of order

Variable Specification Table

Variable	Data Type	Constraint	Purpose	I/O
cName	String	Not null, max 50 characters	Identifies the customer.	Input
cAddress	String	Not null, max 50 characters	Used for delivery if requested	Input
pNumber	String	Not null, max 20 characters	Used for contact purposes	Input
qtyPizza	Integer	Not null, max 6, min 1	Used to calculate the total & discount. Restricted to customers' requirements	Input
sizePizza	String array	Not null, must be (small, medium, large)	Used to calculate the total & discount.	Input

		not case sensitive	Restricted to customers' requirements	
qtyTopping	Integer array	Not null, Min 0	Used to calculate additional cost per pizza	Input
delivery	boolean	Not null	Used to add additional cost to total	Input
subTotal	Float	Calculated cannot be negative	Stores the cost before discount, delivery if selected	Process
discount	Float	Calculated cannot be negative	Stores the discount applied to the order	Process
total	Float	Calculated cannot be negative	Stores the total cost with discount and delivery if selected	Output

Pseudocode

BEGIN

cName = INPUT “Enter your name”

WHILE LENGTH cName < 1 or LENGTH cName >50 AND cName NOT ONLY LETTERS

PRINT “Name must be entered only contain letters and Whitin 1-50"

cName = INPUT “Enter your name”

END WHILE

```
cAddress = INPUT “Enter address”

WHILE LENTH cAddress <1 or LENGTH cAddress  >50
PRINT “Must be between 1-50 to be valid”
cAddress = INPUT “Enter address”
END WHILE

pNumber = INPUT “Enter phone number “
WHILE LENGTH pNumber < 1 OR LENGTH pNumber >20
PRINT “Must be between 1-20 to be valid”
pNumber = INPUT “Enter phone number “
END WHILE

qtyPizza = INPUT “Enter quantity of pizzas “

WHILE qtyPizza  NOT INTERGER OR qtyPizza < 1 OR qtyPizza >6
PRINT “Must be between 1-6 to be valid ”
qtyPizza = INPUT “Enter quantity of pizzas “
END WHILE
sizePizza = LOWERCASE( [])
qtyTopping = LOWERCASE([])


FOR  i =1 TO qtyPizza
sizeRequest   = INPUT “Enter pizza  “ + i +”size”
WHILE LOWERCASE(sizeRequest )  NOT IN [“small”,”medium”,”large”]
PRINT “Pizza must be small, medium or large “
```

```
sizeRequest   = INPUT “Enter pizza  “ + i +”size”
END WHILE
APPEND sizeRequest TO sizePizza
toppingRequest = INPUT “Enter topping amount”
WHILE toppingRequest NOT INTERGER OR toppingRequest <0
PRINT “Must be a whole number Minium of 0”
toppingRequest = INPUT “Enter topping amount”

END WHILE
IF toppingRequest >4
toppingRequest =4
END IF
APPEND toppingRequest TO qtyTopping
END FOR

delivery = INPUT “Enter Yes/No for delivery”
WHILE LENGTH delivery <= 0 OR LOWERCASE(delivery) NOT IN [“yes”,”no”]
delivery = INPUT “Enter Yes/No for delivery”
IF LOWERCASE(delivery) ==  “yes”
delivery = 2.50
END IF

subTotal =0

PRINT “Name” + cName
PRINT “Address” +  cAddress
PRINT “Phone number “ + pNumber
```

```
FOR j =1 TO qtyPizza
IF sizePizza[j] == “small”
PRINT “Small Pizza : £3.25”
subtotal += 3.25
END IF

IF sizePizza[j] == “medium”
PRINT “Medium Pizza : £5.50”
subtotal += 5.50
END IF


IF sizePizza[j] == “Large”
PRINT “Large Pizza : £7.15”
subtotal += 7.15
END IF

IF qtyTopping[j] == 1
PRINT “Topping Amount 1 extra“ +qtyTopping[j] +”£0.75 “
subTotal += 0.75
END IF
IF qtyTopping[j] == 2
PRINT “Topping Amount 2 extra “ +qtyTopping[j] +”£1.35 “
subTotal += 1.35
END IF
IF qtyTopping[j] ==3
PRINT “Topping Amount 3 extra“ +qtyTopping[j] +”£2.0 “
subTotal += 2.0
```

```
END IF

IF qtyTopping[j]  >=4

PRINT “Topping Amount 4 +extra  +“ +qtyTopping[j] +”£ 2.50“

subTotal += 2.50

        END IF

END FOR

PRINT “Subtotal “+  subTotal


IF subTotal > 20

PRINT “Discount applied “ + subTotal *0.10

subTotal - (subtotal *0.10)

PRNT “Delivery charge applied : ” +delivery

IF delivery == 2.50

subTotal += 2.50

PRINT “Total £ “ + subTotal
```

Justification

In this pseudocode I have met all client requirements, throughout this program I have used data validation rules allowing the program to catch errors and resolve it with the user instead of crashing which would have a big impact on the user experience. Further I did not restrict the number of toppings as I felt that it was an ambiguous condition and specific toppings were not specified in the client requirements instead if the user exceeds or enters 4 toppings the program will set the qtyTopping variable to 4. This decision was also made for the longevity of the system and robustness if the client would like to store all orders in a database as it will be more space efficient.

Build

Screenshot	Justification

The rate of one small pizza is :

3.25

The rate of one medium pizza is :

5.5

The rate of one large pizza is :

7.15

The rate of one topping is

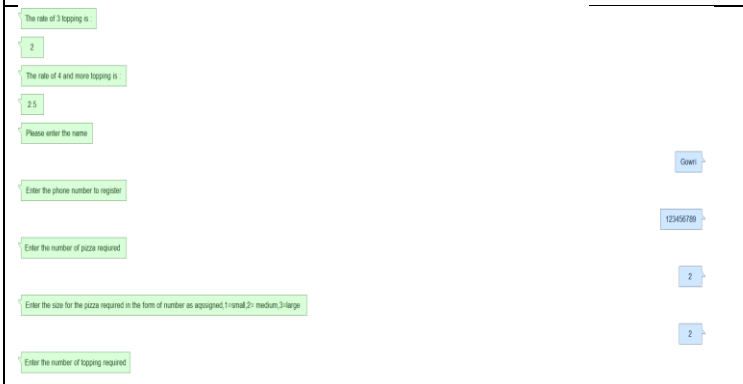
0.75

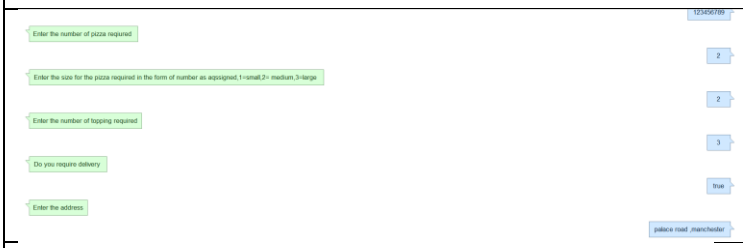
The rate of 2 topping is :

1.35

The rate of 3 topping is :

2





The Customers name is :

Gowri

The Customer address is :

palace road ,manchester

The Customer phone number is :

123456789

The Pizza size ordered:

2

The number of topping ordered:

3

Did the customer choose delivery 1 for yes 0 for no

true

2

The Customer phone number is : 123456789 The Pizza size ordered: 2 The number of topping ordered: 3 Did the customer choose delivery 1 for yes 0 for no true 2 The Discount amount is 0 Your total amount to pay 15.5	

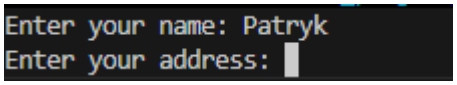
Code test

Test Number	Test purpose	Input data	Expected result	Actual Result	Pass/fail
1	User can input expected name (normal)	Patryk	Program should continue	See Test One	Pass
2	User cannot leave blank (abnormal)	<enter>	Program should wait until acceptable input is given	See Test Two	Pass
3	User can enter alphanumeric For address	12address	Program should continue	See Test Three	Pass

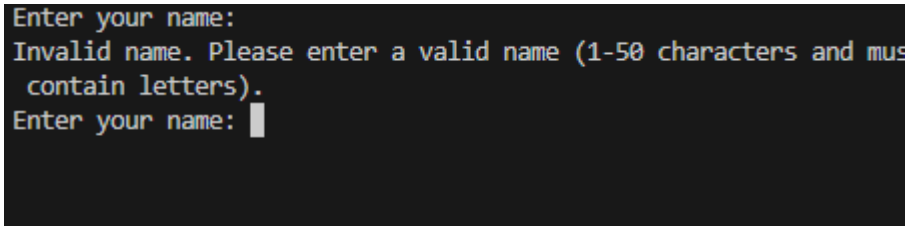
4	Phone Number should not accept negative numbers (Extreme)	-12	Program should wait until acceptable input is given	See Test Four	Pass
5	Quantity of pizza should allow for min and max to be entered (extreme)	1 6	Program should continue	See test five	FAIL
5 Re-test	Quantity of pizza should allow for min and max to be entered (extreme)	1 6	Program should continue	See test 5 retest	Pass
6	Topping accepted One extra topping (extreme)	1	Program should continue	See test 6	Fail
6 Re-test	Topping accepted One extra topping (normal)	1	Program should continue	See test 6 re-test	Pass
7	Topping did not accept characters (abnormal)	one	Program should wait until acceptable input is given	See test 7	Pass (User prompt is not specific enough)
8	Program should calculate correctly For a order without discount and delivery (normal)	Pat 1Address 911 1 small 3 no	Total = £5.25	See test 8	Pass

9	Program should calculate correctly with delivery (normal)	Pat 1Address 911 1 small 3 no	Total = £7.75	See test 9	Pass
10	Program should calculate correctly with discount	Pat 1Address 911 3 Large 2	Subtotal £27.45 Total = £24.71 Discount = £2.75	See test 10	Pass
11	Program should calculate correctly with discount and delivery	Pat 1Address 911 3 Large 2	Subtotal= £27.45 Total = £27.71 Discount = £2.50	See Test 11	Pass

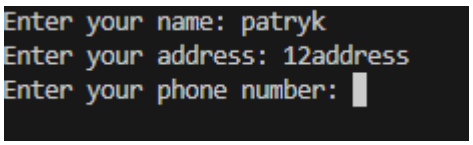
TEST ONE:



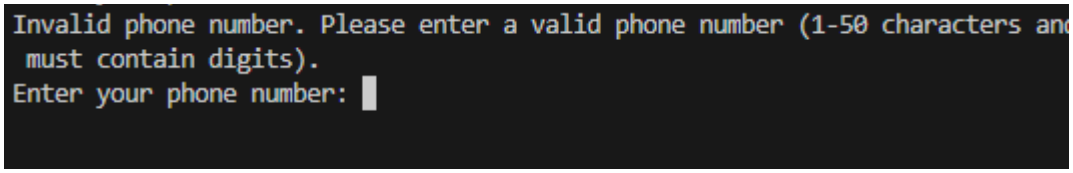
TEST TWO :



TEST THREE:



TEST FOUR:



TEST FIVE:

```
Enter your phone number: 1
Enter the quantity of pizzas (1-6): 1
Do you want the pizza to be delivered? (yes/no): yes
Customer Name: p
 Customer Address: p
 Customer Phone Number: 1

Traceback (most recent call last):
  File "c:\Users\patryk\Documents\[redacted]\mini_project_2_Team
_5.py", line 58, in <module>
    if sizePizza[j] =="small":
       ~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Activate Windows
Go to Settings to activate Win


```
Enter your phone number: 1
Enter the quantity of pizzas (1-6): 1
Do you want the pizza to be delivered? (yes/no): yes
Customer Name: p
 Customer Address: p
 Customer Phone Number: 1

Traceback (most recent call last):
  File "c:\Users\patryk\Documents\[redacted]\mini_project_2_Team
_5.py", line 58, in <module>
    if sizePizza[j] =="small":
       ~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Activate Windows
Go to Settings to activate Win

TEST FIVE CODE :

```
for i in range(0,qtyPizza-1):
```

TEST SIX:

```
Enter your name: p
Enter your address: p
Enter your phone number: 1
Enter the quantity of pizzas (1-6): 1
Enter the size of pizza 1 (small, medium, large):
```

TEST FIVE - RETEST SOLUTION:

In test five when iterating to get the pizza size I had used the number of pizzas the user wanted as a condition however due to logic errors i had set the condition to be one less than entered. This meant when the for-loop check if condition is met it would be set to true as the quantity of the pizza is set to 0 to fix this i removed the minus one.

CODE:

```
for i in range(0,qtyPizza):
```

During test 5 an output was met however the issue is still present bellow is the output with the itemised order list even after stating I would like to order one pizza the flawed logic assumed I had 0. The reason this was not caught when i entered is that this bug happened after validating the input.

```
Enter your name: p
Enter your address: p
Enter your phone number: 1
Enter the quantity of pizzas (1-6): 1
Do you want the pizza to be delivered? (yes/no): yes
Customer Name: p
 Customer Address: p
 Customer Phone Number: 1

Subtotal: £0.00
Delivery Fee: £ 2.50
0
```

Activate Wir
Go to Settings t

TEST SIX :

```
Enter your phone number: 1
Enter the quantity of pizzas (1-6): 1
Enter the size of pizza 0 (small, medium, large): small
Enter the quantity of toppings for pizza 0 Min: 0: 1
Traceback (most recent call last):
  File "c:\Users\patryk\Documents\[redacted]ini_project_2_Team
_5.py", line 44, in <module>
    if toppingRequest >4 :
       ^^^^^^^^^^^^^^^^^
```

TEST SIX CODE:

```
    toppingRequest = input(f"Enter the quantity of toppings
    while not toppingRequest.isdigit() or int(toppingRequest
        print("Invalid quantity. Please enter a non-negative
        toppingRequest = input(f"Enter the quantity of toppi
    if toppingRequest >4 :
        toppingRequest =4
    qtyPizza.append(int(toppingRequest))
```

TEST SIX RE-TEST:

```
Enter the size of pizza 1 (small, medium, large): small
Enter the quantity of toppings for pizza 0 Min: 0: 1
Do you want the pizza to be delivered? (yes/no): no
Customer Name: p
 Customer Address: p
 Customer Phone Number: 1

Pizza 1: Size Small £3.25
Pizza 1: 1 Toppings £0.75
Subtotal: £4.00
Delivery Fee: £0.00
```

CODE FIX:

```

    toppingRequest = input(f"Enter the quantity of toppings
    while not toppingRequest.isdigit() or int(toppingReques
        print("Invalid quantity. Please enter a non-negativ
        toppingRequest = input(f"Enter the quantity of topp
    if int(toppingRequest )>4:
        toppingRequest =4
    qtyTopping.append(int(toppingRequest))
```

When creating toppinRequest I had set it to a string for simplicity instead of using try and excepts to catch an error if the user had entered a string when the data expected is an integer. In doing so any Boolean condition where the integer value is need requires me to recast it to an integer. I had also recaster it to an integer when appending to the list to make calculation easier later on.

TEST SEVEN:

```
Enter the quantity of toppings for pizza 0 Min: 0: one
Invalid quantity. Please enter a non-negative number.
Enter the quantity of toppings for pizza 0 Min: 0:
```

TEST EIGHT :

```
Enter your name: Pat
Enter your address: 1Address
Enter your phone number: 911
Enter the quantity of pizzas (1-6): 1
Enter the size of pizza 1 (small, medium, large): small
Enter the quantity of toppings for pizza 0 Min: 0: 3
Do you want the pizza to be delivered? (yes/no): no
Customer Name: Pat
 Customer Address: 1Address
 Customer Phone Number: 911

Pizza 1: Size Small £3.25
Pizza 1: 3 Toppings £2.00
Subtotal: £5.25
Delivery Fee: £0.00
Total £5.25
```

TEST NINE :

```
Enter your name: Pat
Enter your address: 1Address
Enter your phone number: 911
Enter the quantity of pizzas (1-6): 1
Enter the size of pizza 1 (small, medium, large): small
Enter the quantity of toppings for pizza 0 Min: 0: 3
Do you want the pizza to be delivered? (yes/no): yes
Customer Name: Pat
 Customer Address: 1Address
 Customer Phone Number: 911

Pizza 1: Size Small £3.25
Pizza 1: 3 Toppings £2.00
Subtotal: £5.25
Delivery Fee: £ 2.50
Total £7.75
```

TEST TEN:

```
Customer Name: Pat
 Customer Address: 1Address
 Customer Phone Number: 911

Pizza 1: Size Large £7.15
Pizza 1: 3 Toppings £2.00
Pizza 2: Size Large £7.15
Pizza 2: 3 Toppings £2.00
Pizza 3: Size Large £7.15
Pizza 3: 3 Toppings £2.00
Subtotal: £27.45
Discount applied: £2.75
Total £24.71
```

TEST ELEVEN:

```
Customer Name: Pat
 Customer Address: 1Address
 Customer Phone Number: 911

Pizza 1: Size Large £7.15
Pizza 1: 3 Toppings £2.00
Pizza 2: Size Large £7.15
Pizza 2: 3 Toppings £2.00
Pizza 3: Size Large £7.15
Pizza 3: 3 Toppings £2.00
Subtotal: £27.45
Discount applied: £2.75
Delivery Fee: £ 2.50
Total £27.21
```